

A Technical Introduction to Perplexity language model

Section 1

Token-normalized perplexity In natural language processing (NLP), a corpus is a structured collection of texts or documents, and a language model is a probability distribution over entire texts or documents. Consequently, in NLP, the more commonly used measure is token-normalized perplexity (a token being a word or, more frequently, sub-word), defined as:
$$\left(\prod_{i=1}^N q(s_i) \right)^{-1/N}$$
 where $s_1, \dots, s_n$ are the n documents in the corpus and N is the number of tokens in the corpus. This normalizes the perplexity by the length of the text, allowing for more meaningful comparisons between different texts or models rather than documents. Suppose the average text x_i in the corpus has a probability of 2^{-190} according to the language model. This would give a model perplexity of 2190 for a sentence. However, in NLP, it is more common to normalize by the length of a text. Thus, if the test sample has a length of 1000 tokens, and could be coded using 7.95 bits per token, one could report a model perplexity of $2^{7.95} = 247$. In other words, the model is as confused on test data as if it had to choose uniformly and independently among 247 possibilities for each token. There are two standard evaluation metrics for language models: perplexity or word error rate (WER). The simpler of these measures, WER, is simply the percentage of erroneously recognized words E (deletions, insertions, substitutions) to total number of words N , in a speech recognition task i.e.
$$\text{WER} = \left(\frac{E}{N} \right)$$

Section 2

$$b^{-1/N} \sum_{i=1}^N \log_b q(x_i) = \left(\prod_{i=1}^N q(x_i) \right)^{-1/N},$$

$$\log_b q(x_i) = \left(\prod_{i=1}^N q(x_i) \right)^{-1/N},$$

Section 3

where x ranges over the events, where $0-0$ is defined to be 1, and where the value of b does not affect the result; b can be chosen to be 2, 10, e , or any other positive value other than 1. In some contexts, this measure is also referred to as the (order-1 true) diversity. The logarithm $\log PP(p)$ is the entropy of the distribution; it is given the unit shannon (bit) if the base of the logarithm is 2, and nat if the natural logarithm is used. Perplexity of a random variable X may be defined as the perplexity of the distribution over its possible values x . It can be thought of as a measure of uncertainty or "surprise" related to the outcomes. For a probability distribution p where exactly k outcomes each have a probability of $1/k$ and all other outcomes have a probability of zero, the perplexity of this distribution is

simply k . This is because the distribution models a fair k -sided die, with each of the k outcomes being equally likely. In this context, the perplexity k indicates that there is as much uncertainty as there would be when rolling a fair k -sided die. Even if a random variable has more than k possible outcomes, the perplexity will still be k if the distribution is uniform over k outcomes and zero for the rest. Thus, a random variable with a perplexity of k can be described as being "k-ways perplexed", meaning it has the same level of uncertainty as a fair k -sided die. Perplexity is sometimes used as a measure of the difficulty of a prediction problem. It is, however, generally not a straightforward representation of the relevant probability. For example, if you have two choices, one with probability 0.9, your probability of a correct guess using the optimal strategy is 0.9. Yet, the perplexity is $0.9 - 0.9 \cdot 0.1 - 0.1 = 1.38$. The inverse of the perplexity, $1/1.38 = 0.72$, is not equal to the probability 0.9. The perplexity is the exponentiation of the entropy, a more commonly encountered quantity. Entropy measures the expected or "average" number of bits required to encode the outcome of the random variable using an optimal variable-length code. It can also be regarded as the expected information gain from learning the outcome of the random variable, providing insight into the uncertainty and complexity of the underlying probability distribution.

Section 4

Perplexity of a probability distribution The perplexity PP of a discrete probability distribution p is a concept widely used in information theory, machine learning, and statistical modeling. It is defined as

Section 5

The second metric, perplexity (of a token), is an information theoretic measure that evaluates the similarity of proposed model m to the original distribution p . It can be computed as a inverse of (geometric) average probability of test set T

Section 6

Perplexity of a probability model A model of an unknown probability distribution p may be proposed based on a training sample that was drawn from p . Given a proposed probability model q , one may evaluate q by asking how well it predicts a separate test sample $x_1, x_2, \dots, x_N$ also drawn from p . The perplexity of the model q is defined as

Section 7

$$H(\tilde{p}, q) = - \sum_x \tilde{p}(x) \log_b q(x),$$

$$H(\tilde{p}, q) = - \sum_x \tilde{p}(x) \log_b q(x),$$

Section 8

Recent advances in language modeling Since 2007, significant advancements in language modeling have emerged, particularly with the advent of deep learning

techniques. Token-normalized perplexity, a measure that quantifies the predictive power of a language model, has remained central to evaluating models such as the dominant transformer models like Google's BERT, OpenAI's GPT-4 and other large language models (LLMs). This measure was employed to compare different models on the same dataset and guide the optimization of hyperparameters, although it has been found sensitive to factors such as linguistic features and sentence length. Despite its pivotal role in language model development, perplexity has shown limitations, particularly as an inadequate predictor of speech recognition performance, overfitting and generalization, raising questions about the benefits of blindly optimizing perplexity alone.

Section 9

$$\mathrm{PPL}(D) = \frac{1}{N} \sum_{i=1}^N -\log_2 \left(\frac{1}{m(T_i)} \right),$$

$$\mathrm{PPL}(D) = \sqrt[N]{\prod_{i=1}^N \frac{1}{m(T_i)}} = 2^{\frac{1}{N} \sum_{i=1}^N \log_2(m(T_i))}.$$